MILITARY SPECIFICATION

PUNCH AND DISK, DENTAL RUBBER DAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a rubber dam punch and replacement disk (with screw) suitable for punching holes in dental rubber dams used in dental operative procedures.

1.2 Classification. The components shall be of the following classes:

Class I. - Rubber Dam Punch

Class II. - Replacement Disk (with screw)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RSTS, 2800 South 20th Street, Philadelphia., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

GG-I-526	- Instruments, Dental and Surgical, General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.
ZZ-R-690	- Rubber Dam, Dental.

Military

MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liners, Case, and Sheet; Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	- Metal, Test Methods.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

DRAWING

Defense Personnel Support Center (DPSC)
24526 - Punch and Disk, Dental Rubber Dam

(Copies of specifications standards and drawing required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS are the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM E 92 - Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The punch and disk shall be in accordance with applicable requirements of GG-I-526 and as specified herein.

3.2 Drawing. The drawing forming a part of this specification is a specification control drawing. The supplier shall be responsible for preparing his own shop drawing. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper function, fit, assembly, or operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.3 Material. The punch and disk shall be fabricated of stainless steel conforming to the requirements shown in table I, when tested as specified in 4.4.1.

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3.3.1 Hardness. Hardness shall conform to the requirements shown in table I, when tested as specified in 4.4.2.

TABLE I. - Material (cont't)

Component	Material	Rockwell Hardness
Body	17-4PH Stainless Steel	
Lever	17-4PH Stainless Steel	C40-C46
Disk	440C Stainless Steel	C42-C46
Disk Screw	303 Stainless Steel	
Plunger	440C Stainless Steel	C58-C62
Plunger Screw	416 Stainless Steel	C38-C42
Plunger Spring	FS-302 Stainless Steel	
Hinge Screw	303 Stainless Steel	
Flat Spring	420 Stainless Steel	C50-C55
Detent (Ball)	302 Stainless Steel	
Detent Plug	303 Stainless Steel	
Flat Spring Screw	304 Stainless Steel	

3.4 Design and dimensions. Shall be an adjustable dental rubber dam punch and replacement disk (with screw) suitable for punching round, clean-cut tear-resistant holes in dental rubber dams. Class I shall be furnished as a complete rubber dam punch. Class II shall be furnished as a replacement disk with screw for the rubber dam punch. The design and dimensions of the punch and disk shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 24526.

3.4.1 Locks. Joints of the punch shall be as shown in DPSC Drawing 24526. Locks shall be accurately fitted and without gross crevices, burrs, or sharp edges.

3.5 Cutting performance. The punch shall have a floating plunger design which shall punch dam holes which do not have visually notched or ragged edges even after 20,000 cuts for each hole size. The punch shall be capable of punching holes 2 inches from the edge of the rubber dam without doubling or folding of the dam when tested as specified in 4.4.3.

3.6 Finish. The outer surfaces of the punch shall have a satin, mirror, or polished finish. The outer edge of the disk shall have a straight, 40 pitch knurl. The operative surface of the disk shall be surface ground to produce cutting edges sufficient to meet the requirements specified in 3.5.

3.7 Corrosion resistance. All components of the punch except the disk and plunger shall meet corrosion resistance requirements (copper sulfate and boil test) of GG-I-526, when tested as specified in GG-I-526. FS-302 stainless steel shall be tested as specified in GG-I-526 for type 302 stainless steel.

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3.8 Identification markings. Each punch shall be permanently and legibly marked in accordance with GG-I-526, except as an alternate to the word "stainless", the use of a registered trade name identifying the steel as stainless is acceptable. Punches of foreign manufacture shall also be marked "stainless".

3.9 Workmanship. Cutting edges of each punch shall be properly sharpened and free from burrs or feathered edges. The punch and disk shall be free from defects which detracts from their appearance or impairs their serviceability.

4. . QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II, III, and IV (see 4.2.1, 4.2.2 and 4.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. Unit of product for examination purposes shall be the unit of issue (1 punch).

TABLE II. Sampling for Examination.

Type defect	Inspection Level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be as indicated in Table III.

TABLE III. Sampling for tests.

Component	Characteristic	Requirements	Test Procedure	Unit of Product	Inspection level
Entire Instrument	Hardness	3.3.1	4.4.2	One punch	S-1
Disk & Plunger	Cutting performance	3.5	4.4.3	One punch	*
Entire Instrument **	Copper sulfate test	3.7	GG-I-526	One punch	S-1
Entire Instrument **	Boil test	3.7	GG-I-526	One punch	S-1

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*One sample from the initial sampling lot shall be tested. Failure of sample to pass this test shall be cause for lot rejection.

**Except for disk and plunger.

4.2.3 Raw materials. Sufficient material from each lot of metal used in the construction of the punch and disk shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be tested for the determination of two or more test characteristics.

4.3 Examination. The punch and disk shall be examined for defects including, but not limited to, those listed in table IV.

TABLE IV. Classification of defects.

Categories	Defects
Major A:	
101	Class not as specified.
102	Design of punch or disk not as specified.
103	Disk holes not cleanly cut.
104	Cracks or burrs present on punch or disk.
105	Joint of punch not free from excessive stiffness.
106	Joint of punch has excessive play or is excessively loose.
Major B:	
151	Diameter of disk holes not as specified.
152	Disk does not turn.
Minor A:	
201	Finish of body and lever not as specified.
202	Identification markings not present.
203	Knurling of disk edge not as specified.
Minor B:	
251	Dimensions (other than 151) not as specified.

4.3.1 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.4 Tests.

4.4.1 Chemical analysis. Tests for chemical analysis shall be conducted in accordance with method 111 or 112 of Federal Test Method Standard 151.

4.4.2 Hardness test. Hardness shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

4.4.3 Cutting performance test. The punch shall be tested on rubber dam. Rubber dam shall be of medium thickness conforming to the requirements of Federal Specification ZZ-R-690.

5 PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package.

5.1.1.1.1 Class I. Each rubber dam punch shall be clean, dry and packaged in accordance with MIL-P-116, method IC-2 except that a preservative shall not be used.

5.1.1.1.2 Class II. Each replacement disk with screw shall be clean, dry, and packaged in accordance with MIL-P-116, method IC-1 except that a preservative shall not be used.

5.1.1.4 Intermediate package. Six unit packages containing Class I punches or twelve unit packages containing Class II replacement disks shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for Class I punches or type II disks shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1.

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5.2.1 Level A. Twenty-four intermediate packages containing Class I punches or twelve intermediate packages containing Class II disks shall be packed in an exterior container designed for a Class II load and conforming to PPP-B-601, overseas type or PPP-B-621, Class II. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification.

5.2.2 Level B. Twenty-four intermediate packages containing Class I punches or twelve intermediate packages containing Class II disks shall be packed in an exterior container designed for a Class 2 load and conforming to PPP-B-601, domestic type, PPP-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for Class I punches or Class II disks shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Class I punches or Class II disks packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Class I punches or Class II disks packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

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5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2 and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.3.1.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number,and date of this specification.
(b) Class required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-01-052-6844	PUNCH, DENTAL, RUBBER DAM, Adjustable
6520-01-127-2205	DISK, PUNCH, DENTAL RUBBER DAM

6.3 Subject term (key word) listing:
Punch
Disk

MILITARY INTERESTS:

Custodians:
Army-MD
Navy-MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

AGENT:

DLA-DM

Review Activities:

Army-MD
Navy-MS
Air Force - 03

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Civil Agency Coordinating Activities:
VA-OSS
PHS
FDA-MPQAS

Project No.6520-1895

Location: NCR Disk #2

49024

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-42051	PUNCH AND DISK, DENTAL RUBBER DAM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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